

Biostatistics Assignment: For Applied Human nutrition 2nd year

Due date for submission 25/01/2017 E.C

Total Mark: 15-20%

1. Explain uses and limitation of Biostatistics/statistics
2. Discuss different types of errors in survey
3. Discuss the advantage of sample survey over census
4. Distinguish between Probability and Non-probability sampling
5. Distinguish between stratified and cluster sampling
6. Quota and stratified sampling
7. When 16 seventeen-year-old women were randomly selected and tested with an aerobics test, the mean of their scores was found to be 37.9 with a sample standard deviation of 7.3.
 - A. What is the point estimate of the average Score?
 - B. Construct the 95% confidence interval(use $Z_{\frac{\alpha}{2}}=1.96$)
 - C. Interpret the result
8. Dental association recommends that the time lapse between routine dental checkups should be an average of 6 months. A random sample of 36 patient records at one dental clinic showed the average time between routine checkups to be 9.2 months with S.D of 2 months. Do the sample data indicate the patients wait longer than recommended between dental checkups? Use 5 % level of significance. (Show all your steps)
9. A team of eye surgeons has developed a new technique for a risky eye operation to restore the sight of people blinded from a certain disease. Under the old method it is known that only 30 % of the patients who undergo this operation recover their eye sight. Suppose that surgeons in various hospitals have performed a total of 225 operations using the new method and that 88 have been successful (the patients fully recovered their sights).can we justify the claim that the new method is better than the old one. (Use 1% level of significance)
10. The distribution of diastolic blood pressures for the population of female diabetics between the ages of 30 and 34 has an unknown mean μ and standard deviation $\sigma = 9.1$ mm Hg. It may be useful to physicians to know whether the mean of this population is equal to the

mean diastolic blood pressure of the general population of females in this age group (i.e 74.4mm Hg).

- A. What is the null hypothesis of the appropriate test?
 - B. What is the alternative hypothesis?
 - C. A sample of ten diabetic women is selected; their mean diastolic blood pressure is 84mm Hg. using this information. Conduct a two-sided test at $\alpha = 0.05$ level of significance. What is the p-value of the test'?
 - D. What conclusion do you draw from the results of the test?
 - E. Would your conclusion have been different if you had chosen $\alpha = 0.01$ instead of $\alpha = 0.05$?
11. A doctor claims that about 30 percent of all persons exposed to a certain amount of radiation will feel any ill effects. Among 19 randomly selected samples who were exposed to radiation, only 1 felt any ill effects.
- A. Construct the 95% confidence interval for the population proportion
 - B. Test whether the proportion of population who are exposed to radiation fell any ill effect is less than doctor's claim $\alpha = .05$ level of significance?
 - C. test the null hypothesis $\theta = 0.30$ against the alternative hypothesis $\theta < 0.30$ at the $\alpha = .05$ significance level
12. Suppose that a random sample of 50 bottles of a particular brand of cough medicine is selected and the alcohol content of each bottle is determined. Let μ (μ) denote the average alcohol content for the population of all bottles of the brand under the study. Suppose that the sample mean is 8.2 grams with a standard deviation of 1.5 grams.
- A. Find the 95% confidence interval for the mean alcohol content of the cough medicine. Report the margin of error and show all the work.
 - B. Explain in words any layman can understand what the 95% confidence interval means.
 - C. Would the 90% confidence interval be narrower or wider? Explain why.